This protocol supports two modes of operation.

Active: In this mode it always sends LACP packets/data units (LACPDUs).

Passive: In this mode a node responds only when it receives LACPDUs from a peer. It does not initiate any packet communication.

Generally, the bundled ports are assigned one unique MAC address. If there is a port that is not getting a keep alive packet, it is considered a downlink and will be removed from the LAG. When it is active again, it will be restored in the LAG. This works for both layer 2 and layer 3 switches.

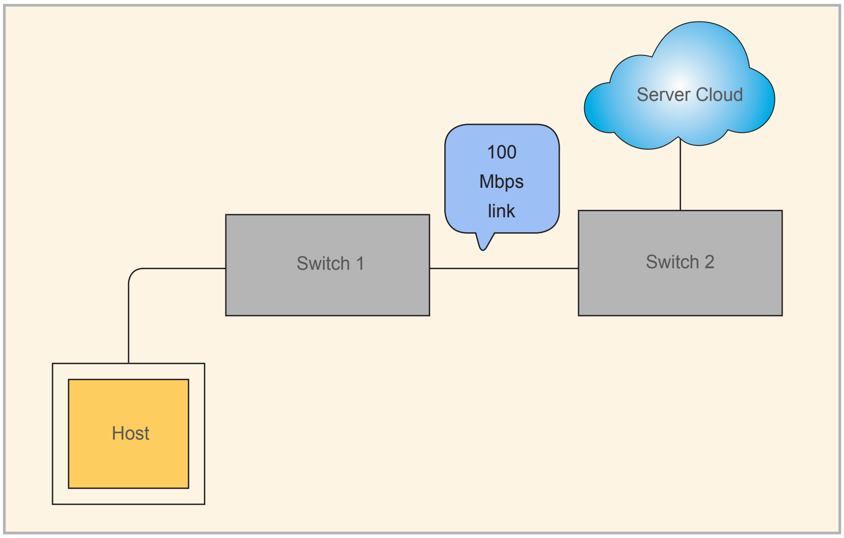

Figure 1: Networking without link aggregation

Handling packets through a group of links

The question now is: How do you load balance the link and send the packets so that they don't arrive out of order from switch1 to switch2? The solution is to create a hash code by XORing a combination of the destination MAC address, IP address and port number. Some implementations only make a string of these three entities and take the lower 3 or 4 bits for making a hash value. For example, in Figure 2, the hash values for five links are stated as:

Link number	*Hash code*
1	001
2	010
3	011
4	100
5	101

For a single process in a destination host, the hash value will always be the same. Thus load balancing will be done for multiple hosts without affecting the packet order.

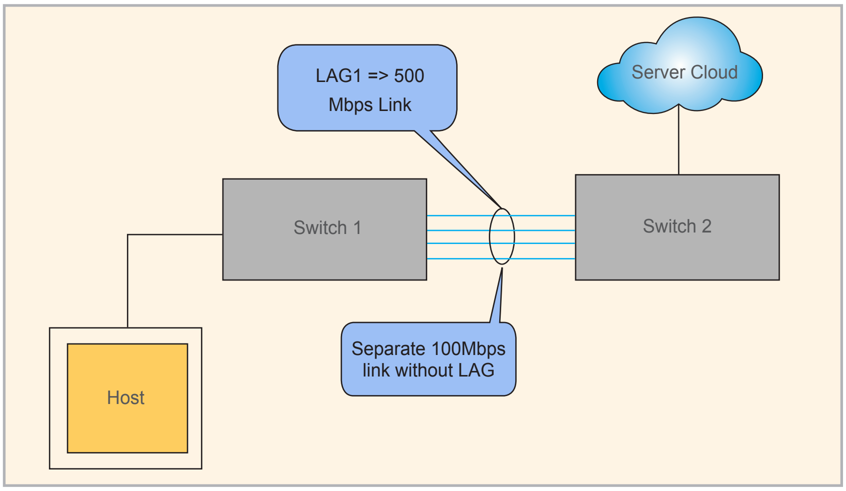

Figure 2: Networking with link aggregation

Linux bonding driver

This driver can be enabled in Linux kernel 2.0 or greater. It provides multiple methods of link aggregation including:

- Active-backup mode
- Round robin mode
- Balanced XOR mode
- LACP protocol

A detailed discussion on these modes is not in the scope of this article and is available in the Linux documentation page.

Limitations of link aggregation

Link aggregation does have a few limitations, however.

- Both the switches should support the same sort of protocol. So inter-compatibility is important.
- All ports of the trunked link should reside on the same switch. They cannot be split into multiple switches. So there is always a chance of a single point of failure.
- All the links should be of the same speed—100Mbps and 1Gbps links cannot be mixed. END

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